

Nyaya Agent: Accelerated Project Plan

IIT Madras | Applied AI & Agentic Systems | 2026 Sprint

1. Project Overview & Schedule

This plan outlines the compressed 3-week development cycle for the Nyaya Agent legal research system. To optimize resources, two developers are active each week, with a final audit performed in the closing stage.

Timeline	Active Personnel	Primary Focus
Week 1	Developer 1 & Developer 2	Foundation: Data Ingestion & Core Agent Logic
Week 2	Developer 3 & Developer 4	Intelligence: Compliance Engine & Graph RAG
Week 3	Developer 5 & Reviewer	Integration: Memory, UI, & Quality Assurance

2. Detailed Roles & Responsibilities

Week 1: The Architects

Developer 1 (Data & Indexing): Build the ingestion pipeline for Indian Kanoon API, SEBI/RBI Circulars, and Gazette of India. Initialize hybrid retrieval (ChromaDB + BM25).

Developer 2 (Agentic Core): Implement LangGraph StateGraph orchestrator to manage agent routing and 3-agent architecture (Research, Compliance, Synthesis).

Week 2: The Domain Specialists

Developer 3 (Compliance Logic): Develop the Compliance Engine for DPDPA/SEBI gap detection. Build tools for Obligation Extraction and Redline Diff generation.

Developer 4 (Knowledge Graph): Construct the Neo4j graph using spaCy NER; map CITES/AMENDED_BY/INTERPRETS relationships to enable Graph RAG.

Week 3: The Integrators

Developer 5 (Full-Stack & Output): Finalize 4-layer memory stack (Window, Summary, SQLite). Develop Streamlit UI with ReAct trace panel and PDF export.

Reviewer (Quality & Validation): Conduct comprehensive code review, perform 100-query load tests, and verify token budget and hallucination controls.

3. Key Milestones

End of Week 1: Demo 1 (Flat RAG) – Functional 3-agent system with basic retrieval.

End of Week 2: Demo 2 (Graph-Comply) – Integrated Compliance Engine and Knowledge Graph.

End of Week 3: Final Delivery – Hardened system with production UI and citation validation.